Artificial Intelligence

Probabilistic Inference

Christopher Simpkins

Kennesaw State University

Approximate Inference for Bayesian Networks

Exact inference in large Bayesian networks is intractable, so we need approximate inference methods. The methods we'll learn are randomized sampling agorithms, a.k.a., **Monte Carlo** algorithms, a.k.a., **stochastic simulation**. They work by

- generating random events based on the probabilities in the Bayes net and
- counting up the different answers found in those random events.

The more the samples, the closer to the true probability distribution.

We'll learn two families of Monte carlo algorithms:

- direct sampling, and
- Markov chain sampling.

Direct Sampling Methods

The primitive element in any sampling algorithm is the generation of samples from a known probability distribution.

Feed a sample from U(1, 1) to the inverse CDF of a distribution to sample the distribution. TODO: example.

This procedure works for a single variable. If we have multiple variables and a Bayes net to represent the joint distribution over them, there are several algorithms

Prior Sampling, a.k.a., Forward Sampling

To sample from a Bayes net with no associated evidence, sample each node in topological order.

- Sample unconditioned nodes according to their prior distributions.
  - Samples become fixed values for sampling CPTs of child nodes.
- Continue sampling child nodes, in order, until all variables are sampled.

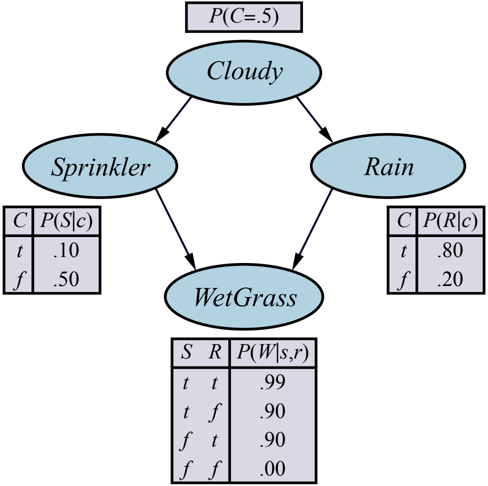
function PRIOR-SAMPLE(*bn*) **returns** an event sampled from the prior specified by *bn*
  inputs: *bn*, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$

  $\mathbf{x} \leftarrow$ an event with n elements
  for each variable X_i **in** $X_1, \ldots, X_n$ **do**
    $\mathbf{x}[i] \leftarrow$ a random sample from $\mathbf{P}(X_i \mid parents(X_i))$
  return $\mathbf{x}$

Example: Generating a Sample from Sprinkler Bayes Net

1. Sample from $P(Cloudy) = \langle 0.5, 0.5 \rangle$
   - Get value of true.
2. Sample from $P(Sprinkler \mid c) = \langle 0.1, 0.9 \rangle$
   - Get value of false.
3. Sample from $P(Rain \mid c) = \langle 0.8, 0.2 \rangle$
   - Get value of true.
4. Sample from $P(WetGrass \mid \neg s, r) = \langle 0.9, 0.1 \rangle$
   - Get value of true.

Full sampled event: $[true, false, true, true]$



From Sampling to Probability Estimates

A Bayes net represents a full joint distribution, so a sample generated from a Bayes net is a sample from the prior joint distribution over all the variables. So, if S_{PS} is a sample generated by the PRIOR-SAMPLE algorithm, then:

$$S_{PS}(x_1, \ldots, x_n) = \prod_{i=1}^{n} Pr\left(x_i \mid parents(X_i)\right) = P(x_1, \ldots, x_n)$$

To estimate these probabilities using samples, we simply generate N samples, count the number of events of interest among the N samples, and divide that number by N. This should converge to the true probability:

$$\lim_{N \to \infty} \frac{N_{PS}(x_1, \ldots, x_n)}{N} = S_{PS}(x_1, \ldots, x_n) = P(x_1, \ldots, x_n)$$

From the CPTs in the Bayes net, the probability of the example event we sampled, $[true, false, true, true]$ is:

$$S_{PS}(true, false, true, true) = 0.5 \cdot 0.9 \cdot 0.8 \cdot 0.9 = 0.324.$$

So if we generated $N = 1000$ samples, we would expect to see $[true, false, true, true]$ 324 times.

Consistent Probability Estimates

An estimate that becomes exact in the large-sample limit is called **consistent**. A consistent estimate of the probability of any partially speciified event $x_1, \ldots, x_m$ for $x \leq n$ is:

$$P(x_1, \ldots, x_m) \approx \frac{N_{PS}(x_1, \ldots, x_m)}{N} = \hat{P}(x_1, \ldots, x_m) \tag{13.6}$$

That is, $\hat{P}(x_1, \ldots, x_m)$ is an approximation of $P(x_1, \ldots, x_m)$ that converges to the true probability as N approaches ∞

- Forward sampling generates *prior* probabilities, that is, probabilities of events prior to observing evidence.
- What we usually want is *posterior* probabilities, $P(X \mid \boldsymbol{e})$ – probability of X given that we have observed evidence $\boldsymbol{e}$.

We now turn to algorithms for computing posteriors, which use prior sampling as a subroutine.

Rejection Sampling

Rejection sampling is a general method for producing samples from a hard-to-sample distribution given an easy-to-sample distribution. In its simplest form, it can be used to compute conditional probabilities, that is, to determine $P(X \mid \boldsymbol{e})$.

Let

- $\hat{\boldsymbol{P}}(X \mid \boldsymbol{e})$ be a vector of the probabilities for each value of $x \in X$ and
- $\boldsymbol{N}_{PS}(X, \boldsymbol{e})$ be a vector of sample counts for each $x \in X$ where the sample is consistent with the evidence $\boldsymbol{e}$ (the variable **C** in the algorithm).

Then:

$$\hat{\boldsymbol{P}}(X \mid \boldsymbol{e}) = \alpha \boldsymbol{N}_{PS}(X, \boldsymbol{e}) = \frac{\boldsymbol{N}_{PS}(X, \boldsymbol{e})}{\boldsymbol{N}_{PS}(\boldsymbol{e})}$$

function REJECTION-SAMPLING(X, **e**, bn, N) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
  inputs: X, the query variable
      e, observed values for variables **E**
      bn, a Bayesian network
      N, the total number of samples to be generated
  local variables: **C**, a vector of counts for each value of X, initially zero

  for j = 1 **to** N **do**
    x ← PRIOR-SAMPLE(bn)
    if x is consistent with **e then**
      C[j] ← **C**[j]+1 where x_j is the value of X in **x**
  return NORMALIZE(**C**)

Rejection Sampling Example

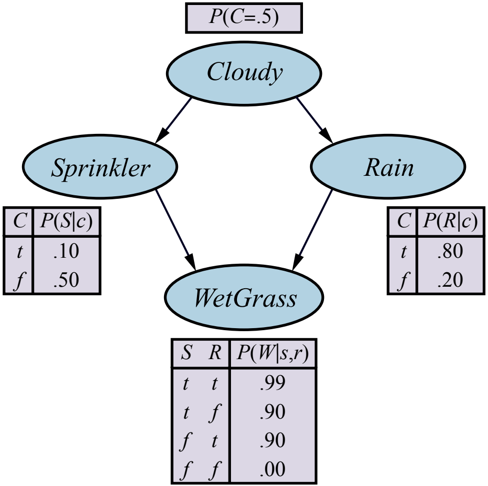
Say we use prior sampling to generate 100 samples and we want to estimate $\boldsymbol{P}(Rain \mid Sprinkler = true)$.

- 73 samples have $Sprinkler = false$, so are *rejected*.
- Of the 27 *accepted* samples:
  - 8 samples have $Rain = true$
  - 19 samples have $Rain = false$.

Then:

$$
\begin{aligned}
\hat{\boldsymbol{P}}(Rain \mid Sprinkler = true) &= \frac{\boldsymbol{N}_{PS}(X, \boldsymbol{e})}{\boldsymbol{N}_{PS}(\boldsymbol{e})} \\
&= \frac{\langle 8, 19 \rangle}{27} \\
&= \langle 0.296, 0.704 \rangle
\end{aligned}
$$

Calculating directly from the CPTs in the Bayes net would give us $\langle 0.3, 0.7 \rangle$.



Importance Sampling

The general statistical technique of **importance sampling** aims to emulate the effect of sampling from a distribution P using samples from another distribution Q whose samples are easier to obtain.

We ensure that the answers are correct in the limit by applying a correction factor $\frac{P(\boldsymbol{x})}{Q(\boldsymbol{x})}$, also known as a **weight**, to each sample $\boldsymbol{x}$ when counting up the samples.

Our query variable will always be one of the nonevidence variables, Z. The idea of importance sampling is to sample from $P(\boldsymbol{z} \mid \boldsymbol{e})$:

$$\hat{P}(\boldsymbol{z} \mid \boldsymbol{e}) = \frac{N_P(\boldsymbol{z})}{N} \approx P(\boldsymbol{z} \mid \boldsymbol{e})$$

but using an arbitrary distribution $Q(\boldsymbol{z})$:

$$\hat{P}(\boldsymbol{z} \mid \boldsymbol{e}) = \frac{N_Q(\boldsymbol{z})}{N} \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q(\boldsymbol{z})} \approx Q(\boldsymbol{z}) \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q(\boldsymbol{z})} = P(\boldsymbol{z} \mid \boldsymbol{e})$$

The question is: which Q to use? We want Q as close as possible to $P(\boldsymbol{z} \mid \boldsymbol{e})$ with $Q(\boldsymbol{z}) \neq 0$ whenever $P(\boldsymbol{z} \mid \boldsymbol{e}) \neq 0$. The most common approach is **likelihood weighting** . . .

Likelihood Weighting

Fix the values for the evidence variables $\boldsymbol{E}$ and sample all the nonevidence variables in topological order, each conditioned on its parents – guarantees each sample consistent with evidence.

Let the sampling distribution be Q_{WS} and nonevidence variables be $\boldsymbol{Z} = \{Z_1, \ldots, Z_l\}$. Then:

$$Q_{WS}(\boldsymbol{z}) = \prod_{i=1}^{l} P(z_i \mid parents(Z_i))$$

Now we need to know the weight for each sample. By general scheme for importance sampling:

$$w(\boldsymbol{z}) = \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})} = \alpha \frac{P(\boldsymbol{z}, \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})}$$

where the normalizing factor $\alpha = \frac{1}{P(e)}$ is the same for all samples. Since $\boldsymbol{z}$ and $\boldsymbol{e}$ cover all the variables in the Bayes net, $P(\boldsymbol{z}, \boldsymbol{e})$ is the product of the conditional probabilities for the nonevidence variables times the product of the conditional probabilities for the evidence variables:

$$\begin{aligned} w(\boldsymbol{z}) &= \alpha \frac{P(\boldsymbol{z}, \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})} = \alpha \frac{\prod_{i=1}^{l} P(z_i \mid parents(Z_i)) \prod_{i=1}^{m} P(e_i \mid parents(E_i))}{\prod_{i=1}^{l} P(z_i \mid parents(Z_i))} \\ &= \alpha \prod_{i=1}^{m} P(e_i \mid parents(E_i)) \qquad (14.9) \end{aligned}$$

The name of this method comes from the fact that probabilities of evidence are generally called likelihoods.

Likelihood Weighting Algorithm

function LIKELIHOOD-WEIGHTING($X, \mathbf{e}, bn, N$) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
  inputs: X, the query variable
    $\mathbf{e}$, observed values for variables $\mathbf{E}$
    bn, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$
    N, the total number of samples to be generated
  local variables: $\mathbf{W}$, a vector of weighted counts for each value of X, initially zero

  for j = 1 **to** N **do**
    $\mathbf{x}, w \leftarrow$ WEIGHTED-SAMPLE($bn, \mathbf{e}$)
    $\mathbf{W}[j] \leftarrow \mathbf{W}[j] + w$ where x_j is the value of X in $\mathbf{x}$
  return NORMALIZE($\mathbf{W}$)

function WEIGHTED-SAMPLE($bn, \mathbf{e}$) **returns** an event and a weight

  $w \leftarrow 1$; $\mathbf{x} \leftarrow$ an event with n elements, with values fixed from $\mathbf{e}$
  for i = 1 **to** n **do**
    if X_i is an evidence variable with value x_{ij} in $\mathbf{e}$
      then $w \leftarrow w \times P(X_i = x_{ij} \mid parents(X_i))$
      else $\mathbf{x}[i] \leftarrow$ a random sample from $\mathbf{P}(X_i \mid parents(X_i))$
  return $\mathbf{x}$, w

Likelihood Weighting Example

Given the query

- $P(Rain \mid Cloudy = true, WetGrass = true)$

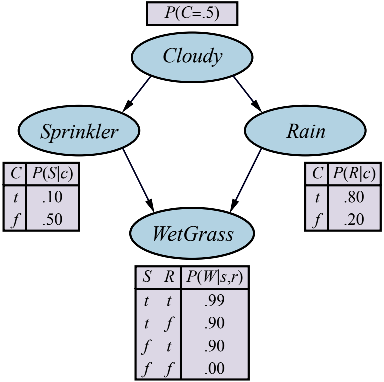
and the ordering

- $Cloudy, Sprinkler, Rain, WetGrass,$

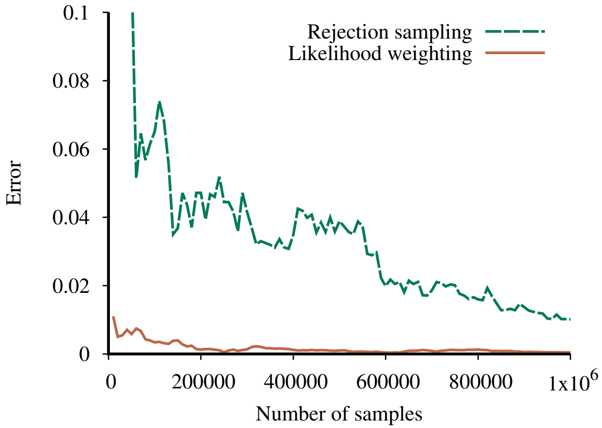
we initialize $w = 1.0$ and proceed as follows:

1. $Cloudy$ is an evidence variable with value true. Therefore, we set
$$w \leftarrow w \cdot P(c) = 0.5.$$
2. $Sprinkler$ is not an evidence variable, so sample from $P(Sprinkler \mid Cloudy = true) = \langle 0.1, 0.9 \rangle$; suppose this returns false.
3. $Rain$ is not an evidence variable, so sample from $P(Rain \mid Cloudy = true) = \langle 0.8, 0.2 \rangle$; suppose this returns true.
4. $WetGrass$ is an evidence variable with value true. Therefore, we set
$$w \leftarrow w \times P(WetGrass = true \mid \neg s, r) = 0.5 \cdot 0.9 = 0.45.$$

Here WEIGHTED-SAMPLE returns the event $[true, false, true, true]$ with weight 0.45, which we tally under $Rain = true$.



Rejection vs. Importance Sampling



Markov Chain Monte Carlo (MCMC) Algorithms

Instead of generating each sample from scratch, MCMC algorithms generate a sample by making a random change to the preceding sample.

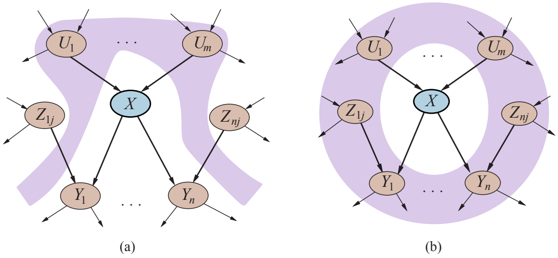
Think of an MCMC algorithm as being in a particular current state that specifies a value for every variable and generating a next state by making random changes to the current state.

We'll learn two MCMC algorithms:

- Gibbs sampling, and
- Metropolis-Hastings

Gibbs Sampling and Markov Blankets

Conditional Independence Relations

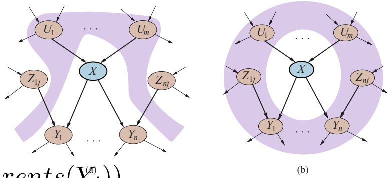


- (a) Each variable is conditionally independent of its non-descendants, given its parents.
- (b) A variable is conditionally independent of all other nodes in the network, given its parents, children, and children's parents – that is, given its **Markov blanket**.

For example, the variable $Burglary$ is independent of $JohnCalls$ and $MaryCalls$, given $Alarm$ and $Earthquake$.

Gibbs Sampling

- Fix the evidence variables.
- Start in a arbitrary state sampled from the nonevidence variables.
  - Each X_i is sampled from the values in its Markov blanket, $mb(X_i)$



$$P(x_i \mid mb(X_i)) = \alpha P(x_i \mid parents(X_i)) \prod_{Y_j \in Children(X_i)} P(y_i \mid parents(Y_j)) \qquad (13.10)$$

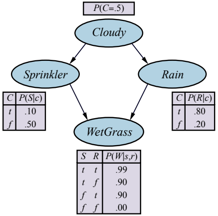
- Wander the state space randomly, keeping the evidence variables fixed and flipping one nonevidence variable by sampling from a distribution calculated Equation 13.10.

Gibbs sampling for X_i means sampling conditioned on the current values of the variables in its Markov blanket.

Gibbs Sampling Example: Step 1

Consider the query $P(Rain \mid Sprinkler = true, WetGrass = true)$.

- Evidence variables $Sprinkler$ and $WetGrass$ fixed to observed values (true), nonevidence variables $Cloudy$ and $Rain$ initialized randomly to, say, true and false respectively.
  - Initial state is $[true, \textbf{true}, false, \textbf{true}]$. Fixed evidence variables marked in bold.



Now the nonevidence variables Z_i are sampled repeatedly in some random order according to a probability distribution $\rho(i)$ for choosing variables. For example:

1. $Cloudy$ is chosen and then sampled, given the current values of its Markov blanket: in this case, we sample from $P(Cloudy|Sprinkler = true, Rain = false)$ whose distribution is calculated using Equation 13.10:

$$P(x_i \mid mb(X_i)) = \alpha P(x_i \mid parents(X_i)) \prod_{Y_j \in Children(X_i)} P(y_i \mid parents(Y_j)) \qquad (13.10)$$

$$P(c \mid s, \neg r) = \alpha P(c) P(s \mid c) P(\neg r \mid c) = \alpha 0.5 \cdot 0.1 \cdot 0.2$$

$$P(\neg c \mid s, \neg r) = \alpha P(\neg c) P(s \mid \neg c) P(\neg r \mid \neg c) = \alpha 0.5 \cdot 0.5 \cdot 0.8$$

yielding $\alpha \langle 0.001, 0.020 \rangle \approx \langle 0.048, 0.952 \rangle$

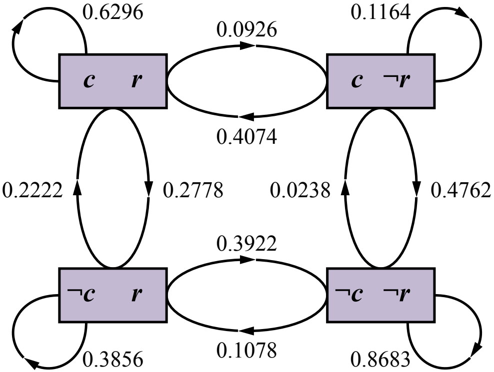
- Suppose the result is $Cloudy = false$. Then the new current state is $[false, \textbf{true}, false, \textbf{true}]$.

Gibbs Sampling Example: Step 2

2. Sampling $\rho(i)$ then chooses $Rain$ as next variable. We sample from $P(Rain \mid Cloudy = false, Sprinkler = true, WetGrass = true)$ using Equation 13.10 (just like Step 1, not shown here).
   - Suppose this yields $Rain = true$. The new current state is $[false, \textbf{true}, true, \textbf{true}]$.

Figure on the right shows the complete Markov chain for the case where variables are chosen uniformly, i.e., $\rho(Cloudy) = \rho(Rain) = 0.5$.

- The algorithm wanders around in this graph, following links with stated probabilities.
- Each state visited during this process is a sample that contributes to the estimate for the query variable Rain.
  - If the process visits 20 states where Rain is true and 60 states where Rain is false, then the answer to the query is $NORMALIZE(\langle 20, 60 \rangle) = \langle 0.25, 0.75 \rangle$.



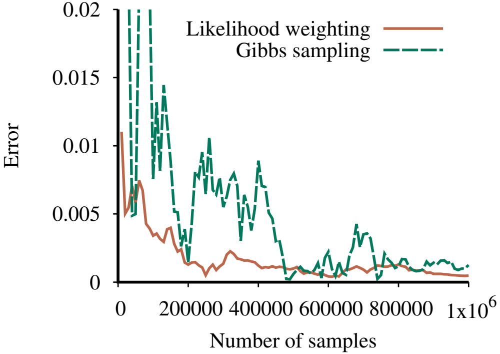
Gibbs Sampling Algorithm

function GIBBS-ASK($X, \mathbf{e}, bn, N$) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
local variables: $\mathbf{C}$, a vector of counts for each value of X, initially zero
$\mathbf{Z}$, the nonevidence variables in bn
$\mathbf{x}$, the current state of the network, initialized from $\mathbf{e}$

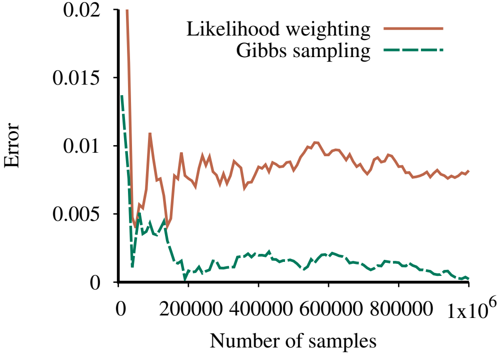
initialize $\mathbf{x}$ with random values for the variables in $\mathbf{Z}$
for $k = 1$ **to** N **do**
choose any variable Z_i from $\mathbf{Z}$ according to any distribution $\rho(i)$
set the value of Z_i in $\mathbf{x}$ by sampling from $\mathbf{P}(Z_i \mid mb(Z_i))$
$\mathbf{C}[j] \leftarrow \mathbf{C}[j] + 1$ where x_j is the value of X in $\mathbf{x}$
return NORMALIZE($\mathbf{C}$)

Gibbs Sampling vs. Importance Sampling

Performance of Gibbs sampling compared to likelihood weighting on the car insurance network.



(a)

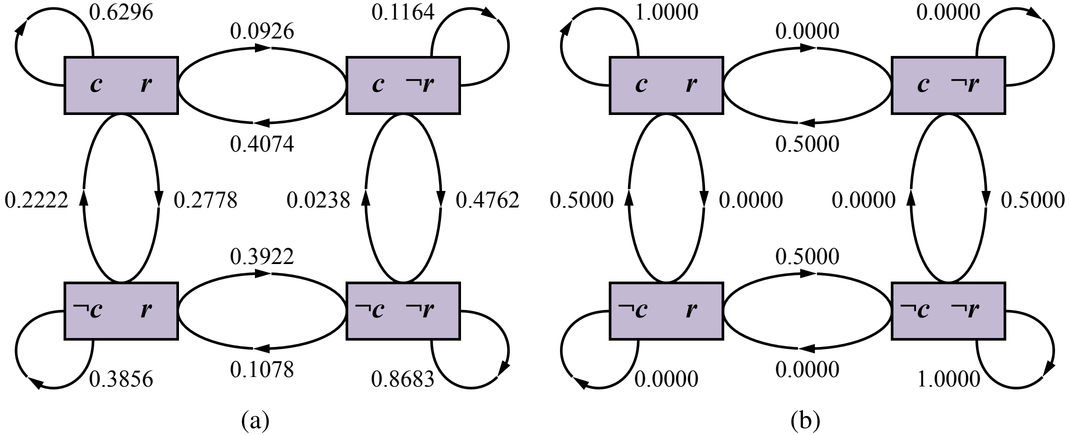


(b)

- (a) for the standard query on $PropertyCost$, and
- (b) for the case where the output variables are observed and Age is the query variable.

Gibbs sampling typically outperforms likelihood weighting when evidence is mostly downstream.

Markov Chains



- (a) The states and transition probabilities of the Markov chain for the query $P(Rain|Sprinkler = true, WetGrass = true)$. Note the self-loops: the state stays the same when either variable is chosen and then resamples the same value it already has.
- (b) The transition probabilities when the CPT for $Rain$ constrains it to have the same value as $Cloudy$.

Metropolis


METROPOLIS
MANUSKRIPT: THEA VON HARBOU
MUSIK: GOTTFRIED HUPPERTZ
EIN FILM VON FRITZ LANG
JN DEN HAUPTROLLEN:
BRIGITTE HELM · GUSTAV FRÖHLICH.
ALFRED ABEL, RUDOLF KLEIN-ROGGE, THEODOR LOOS, FRITZ RASP, HEINRICH GEORGE
AN DER KAMERA: KARL FREUND, GÜNTHER RITTAU
UFA FILM IM VERLEIH DER PARUFAMET

Metropolis-Hastings Sampling

Like simulated annealing, MH has two stages in each iteration of the sampling process:

1. Sample a new state x' from a **proposal distribution** $q(x'|x)$, given the current state x.
2. Probabilistically accept or reject x' according to the **acceptance probability**

$$a(x'|x) = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right)$$

If the proposal is rejected, the state remains at x.

$q(x'|x)$ could be defined as follows:

- With probability 0.95, perform a Gibbs sampling step to generate x'.
- Otherwise, generate x' by running WEIGHTED-SAMPLE using likelihood weighting.

This has the effect of getting MH "unstuck."